

HVAC System Optimization Report

Ottawa Commercial Building Case Study

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Executive Summary

Baseline Annual Energy : 662,256 kWh
Baseline Annual Cost : \$79,470.72

Best Scenario: Combined Measures

- Energy Reduction : 85,410 kWh (12.9%)
- Cost Savings : \$10,249.00
- Payback Period : 4.9 years
- CO₂ Reduction : 42.7 tonnes

Parametric Optimization found even better configuration saving an extra ~\$9,840/year.

Monte Carlo Simulation - ROI Distribution (5000 runs)

